

HHardwareHub

ASSESSMENT 100% COMPLETE

Full-Stack Product Management Suite

Intern Assessment Report: Laravel 12 API + React Web + Flutter Mobile

BACKEND
Laravel 12 / Sanctum

WEB FRONTEND
React 18 + Vite

MOBILE APP
Flutter 3.x (Android/iOS)

DATABASE
MySQL 8.0 (XAMPP)

1 Executive Summary & Architecture

This project implements an enterprise-grade Product & Inventory Suite in exact compliance with the assessment guidelines. It connects a high-performance **Laravel 12 REST API Gateway**, a responsive **React Web Portal**, and a production-grade **Flutter Mobile App** equipped with an automatic multi-URL fallback engine for zero-config physical device connectivity.

 **Laravel 12 Backend**
Sanctum Token Authentication, RESTful endpoints, soft delete recovery, and transactional stock calculation.

 **React 18 Web App**
Tailwind CSS dark-header tokens, real-time KPI metrics, Product CRUD, and one-click CSV/PDF export.

 **Flutter Mobile App**
Custom Slate/Indigo design tokens, unblocked inline subheader action pills, and auto-detect USB/LAN engine.

2 Assessment Requirement Compliance Matrix

PDF REQUIREMENT AREA	SPECIFIED CRITERIA	IMPLEMENTATION DETAILS	STATUS
1. Laravel Application	Login & Logout Authentication	Sanctum token & session auth with secure validation	✓ Completed
	Authenticated Dashboard	Real-time KPIs: Total Products, Units, Valuation, Low Stock	✓ Completed
	Users Management Page	Registered users list with role badges & timestamps	✓ Completed
	Product Management Page	Create, View/List, Edit, and Delete with Laravel validation	✓ Completed
2. Product Data Schema	SKU, Name, Desc, Price, Qty, Status	Extended with Category, Cost Price, and Min Stock Alerts	✓ Completed
3. Soft Delete & Recycle Bin	Soft Deletes via deleted_at	Products and stock movements use SoftDeletes trait	✓ Completed
	Recycle Bin Screen	Dedicated tab listing deleted records	✓ Completed
	Restore & Permanent Delete	1-Click instant recovery & permanent purge options	✓ Completed
4. Laravel REST API	Mobile Login / Logout APIs	Sanctum Bearer token generation & revocation	✓ Completed
	Product CRUD & Recycle APIs	Full standard REST endpoints with route protection	✓ Completed
5. Flutter Mobile App	Login, Dashboard, CRUD Screens	Modern multi-tab navigation, custom cards, form dialogs	✓ Completed
	Mobile Recycle Bin Screen	Dual-tab recovery for products & stock ledger entries	✓ Completed
Bonus Extensions	Stock Ledger Audit Trail	IN, OUT, ADJUSTMENT, DAMAGE balance tracking	★ Bonus Added
	Multi-URL Auto Fallback Engine	Auto-connects USB (127.0.0.1) & Wi-Fi LAN (192.168.1.3)	★ Bonus Added

3 Database Schema & API Reference

📦 Products Table (`products`)

FIELD	TYPE	DESCRIPTION
id	BIGINT	PK, Auto Increment
sku	VARCHAR	Unique Product SKU
name	VARCHAR	Product Name
category	VARCHAR	Category Tag
selling_price	DECIMAL	Selling Unit Price
stock_quantity	INT	Real-time Units
status	ENUM	active / inactive
deleted_at	TIMESTAMP	Soft Delete Flag

📦 Stock Movements Table (`stock\_movements`)

FIELD	TYPE	DESCRIPTION
id	BIGINT	PK, Auto Increment
product_id	BIGINT	FK to `products`
type	ENUM	in, out, adj, dmg
quantity	INT	Quantity Changed
previous_stock	INT	Audit Trail Start
new_stock	INT	Audit Trail End
reason	VARCHAR	Movement Reason
deleted_at	TIMESTAMP	Soft Delete Flag

🌐 Key REST API Endpoints (`/api/v1`)

METHOD	ENDPOINT	DESCRIPTION	ACCESS
POST	/auth/login	Authenticate user & issue Sanctum Bearer token	Public
GET	/dashboard/summary	Retrieve live KPI metrics & count summaries	Sanctum
GET/POST	/products	List (with search/filters) or create product SKU	Sanctum
GET/PUT/DELETE	/products/{id}	View, update, or soft-delete product	Sanctum
GET/POST/DELETE	/stock-movements	Audit log, record stock movement, or void	Sanctum
GET/POST/DELETE	/recycle-bin/products	List soft-deleted items, restore, or permanent purge	Sanctum

4 Step-by-Step Execution Guide

🔑 **Default Admin Credentials:** Email: admin@hardwarehub.com | Password: password

Step 1: Database & Backend Server

1. Start **MySQL** in XAMPP (Port 3306).
2. Open Terminal 1 and start Laravel server:

```
cd d:\HardwareHub\hardwarehub-backend
php artisan serve --host 0.0.0.0 --port 8000
```

Step 2: React Web Frontend

1. Open Terminal 2 and start Vite server:

```
cd d:\HardwareHub\hardwarehub-backend
npm run dev
```

2. Open browser: http://localhost:5173

Step 3: Flutter Mobile Application

Connect your Android phone (Samsung Galaxy A05s) via USB or Wi-Fi, then run:

```
& "$env:LOCALAPPDATA\Android\Sdk\platform-tools\adb.exe" reverse tcp:8000 tcp:8000
cd d:\HardwareHub\hardwarehub_mobile
flutter run -d R7AW903PBSM
```

* Note: The app automatically detects USB mode (127.0.0.1:8000) and Wi-Fi mode (192.168.1.3:8000) via its built-in Auto-Fallback Engine.